Operating System Security

Lab 10

Using UFW and SSH

Alex MacIntyre

Requirements:

Installing and configuring the Ubuntu Firewall

- **Install the firewall & confirmed it is installed (Yes you may already have it installed please show how to do the process)**
- **Show the firewall is enabled and running**
- **Show how to see what ports are open and what ports are listening**

Using SSH

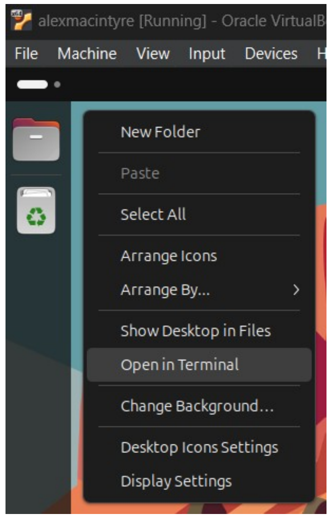
- **Install SSH:**
- **Install PUTTY:**
- **Create the connection:**
- **Once complete, go change the ssh port in the config file**
- **Use Putty to show the connection now working with the new ssh port**
- **Change the ssh port back port 22**

Scripting: Passing Command Arguments/Parameters to a Script

- **Create a script and then run it to pass the parameters to it**
- **IPV4**
- **Subnet Mask**
- **Default Gateway**
- **Then display all three parameters on the same line: "Config updates are:"**
- **Show the Script Code**
- **Run the Script and show the results passing the configurations**

Part I – Installing and Configuring the Ubuntu Firewall

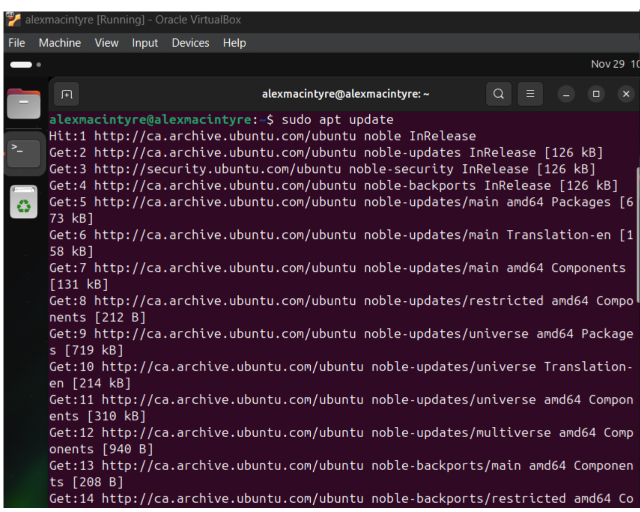
Step One – Open the terminal. This can be done by pressing "**CTRL + ALT + T**" on the keyboard, or by right clicking the desktop and pressing "**Open in Terminal**".

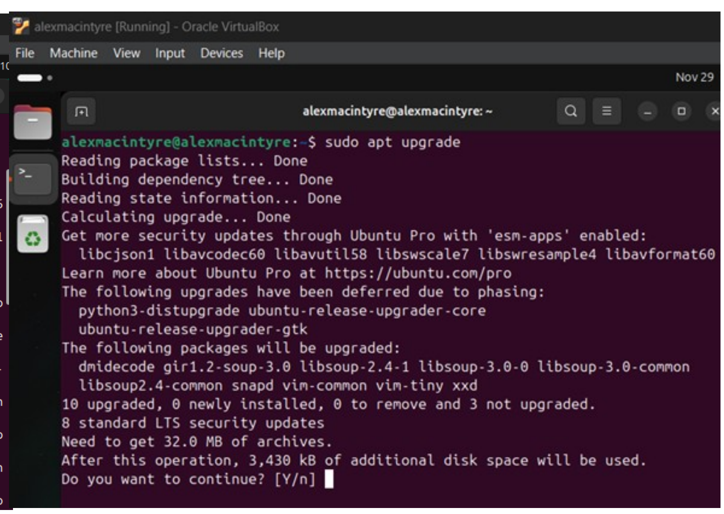


Step Two - Once the terminal is open, ensure that Ubuntu is fully updated and upgraded.

To update Ubuntu, type "**sudo apt update**" into the terminal and press **Enter**.

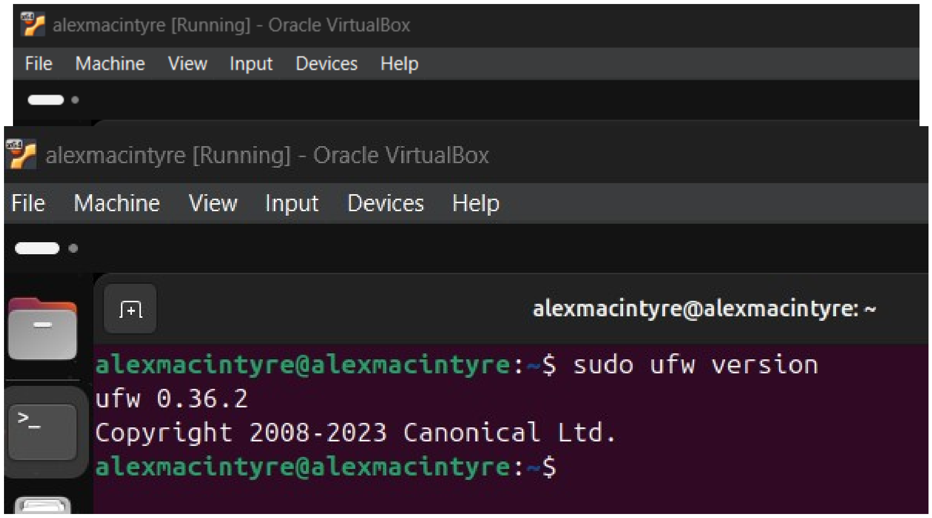
To upgrade Ubuntu, type "**sudo apt upgrade**" and press **Enter**. There will be a prompt to complete the operation. Type "**Y**" and press **Enter.**



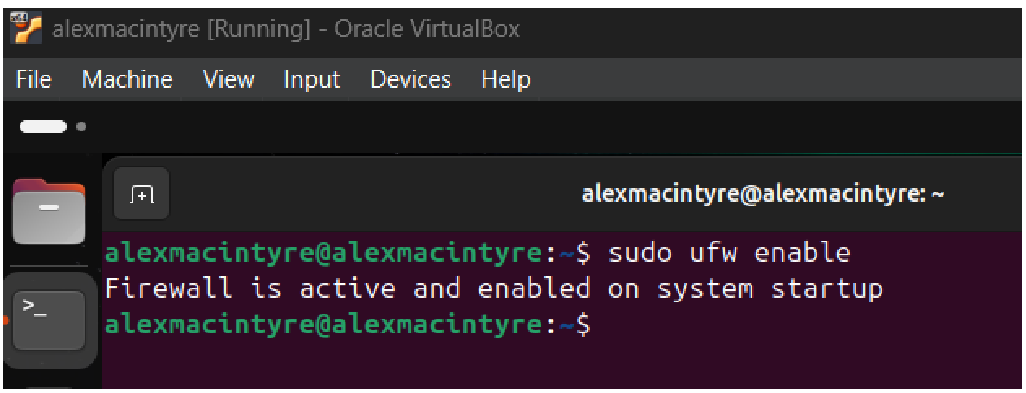


Step Three – Install UFW by typing "**sudo apt install ufw**" and pressing **Enter.**

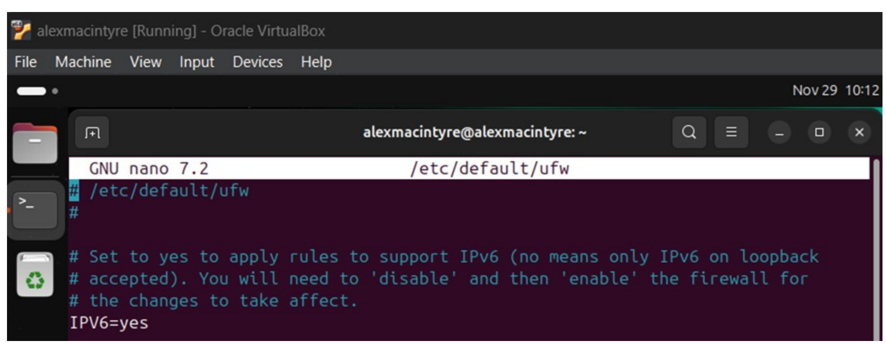
Step Four – Confirm that UFW was successfully installed by typing "**sudo ufw version**".

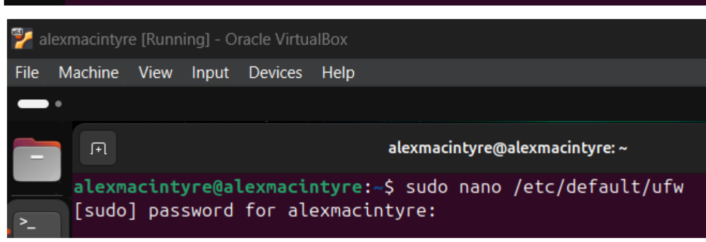


Step Five – Enable the firewall by typing "**sudo ufw enable**" and pressing **Enter**.

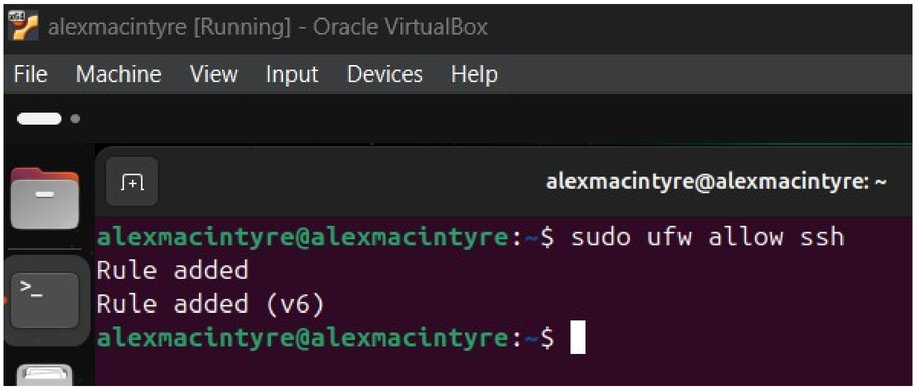


Step Six – To verify the IPv6 status, type "**sudo nano /etc/default/ufw**" and press **Enter** to open the Text Editor. Enter the **sudo password** and press **Enter**. When the Text Editor is open, IPv6 should say "**yes**" next to it. If it doesn't, change it to "**yes**" and press "**CTRL+X**" to save and exit.

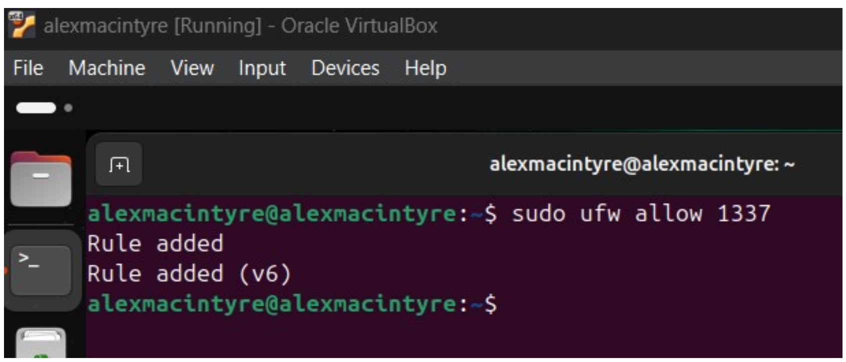




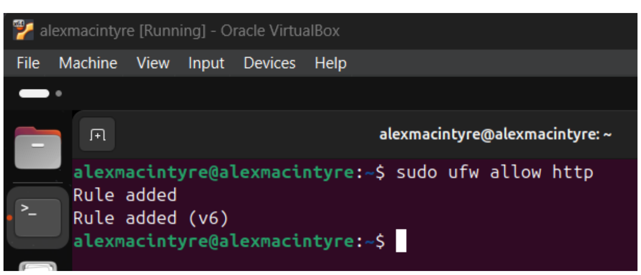
Step Seven – Add a rule to the firewall that allows SSH. Type “**sudo ufw allow ssh**” and press **Enter**.

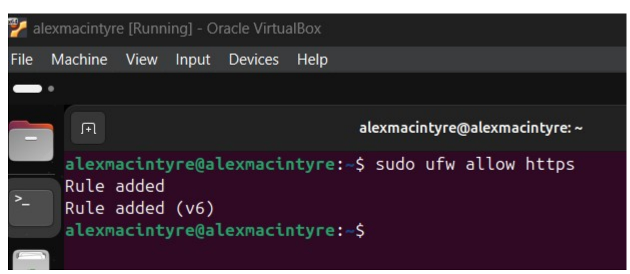


Step Eight – Add a rule to the firewall that enables port number 1337.

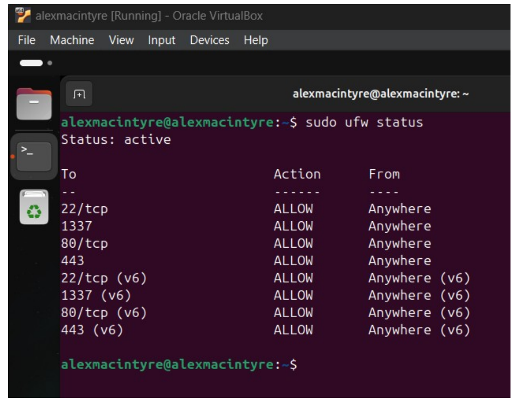


Step Nine - Add a rule to the firewall that allows HTTP and HTTPS connections. For HTTP, type “**sudo ufw allow http**” and press **Enter**. For HTTPS, type “**sudo ufw allow https**” and press **Enter**.

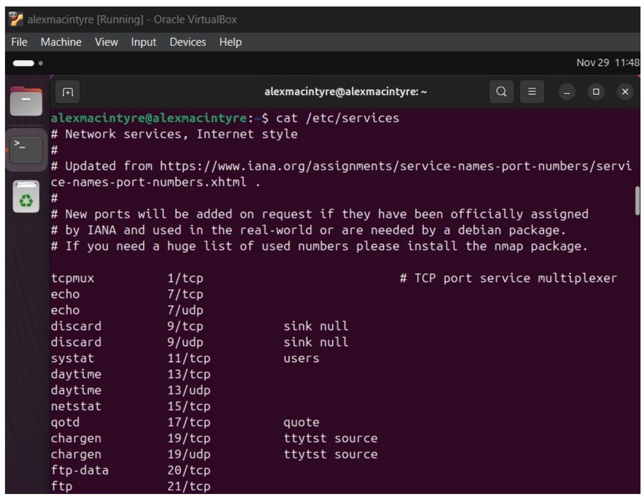




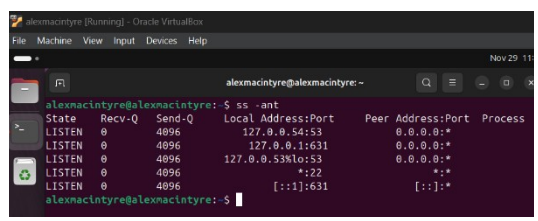
Step Ten – To check the status of the firewall and rule configurations, type "**sudo ufw status**" and press **Enter.**



Step Eleven – To view **Network Services**, which includes a list of ports that are in use, type "**cat /etc/services**" and press **Enter**.



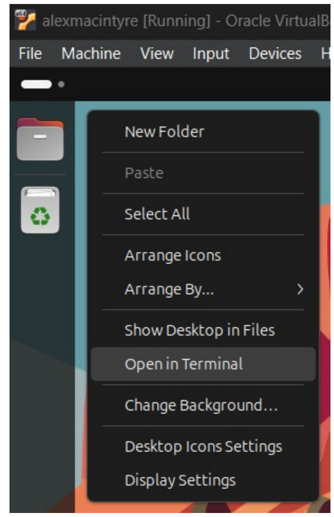
Step Twelve – To see which ports are listening, type **"ss -ant"** and press **Enter.**



Part II – Using SSH

Installing OpenSSH on Linux

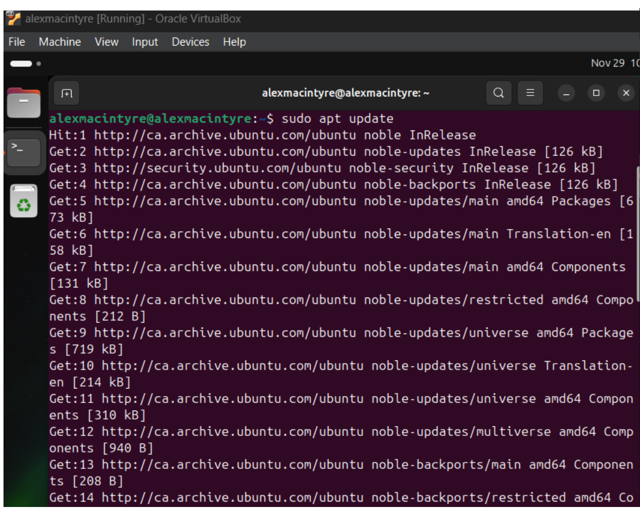
Step One – Open the terminal. This can be done by pressing "**CTRL + ALT + T**" on the keyboard, or by right clicking the desktop and pressing "**Open in Terminal**".

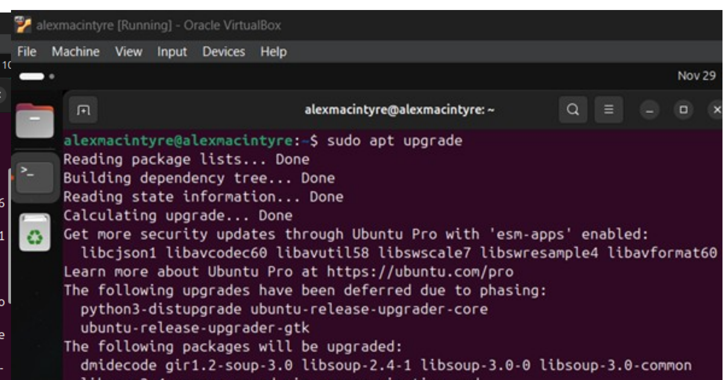


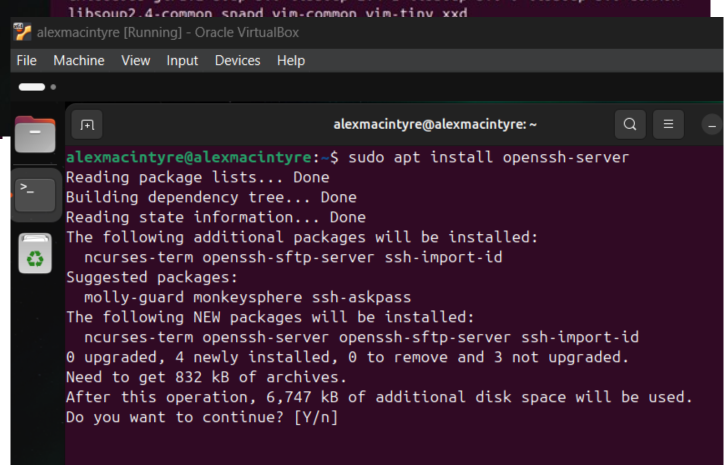
Step Two – Once the terminal is open, ensure that Ubuntu is fully updated and upgraded.

To update Ubuntu, type "**sudo apt update**" into the terminal and press **Enter**.

To upgrade Ubuntu, type "**sudo apt upgrade**" and press **Enter**. There will be a prompt to complete the operation. Type "**Y**" and press **Enter.**

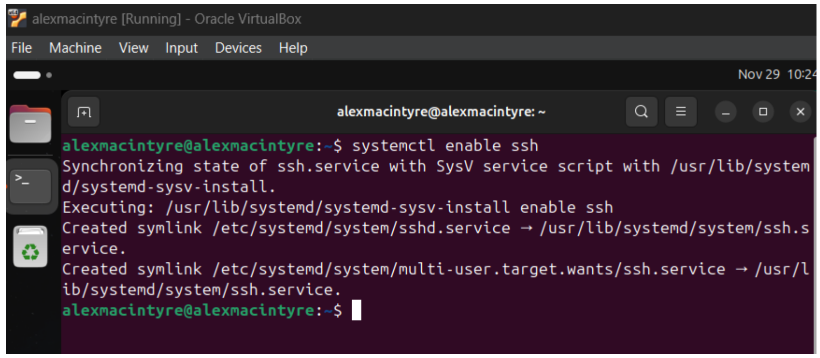


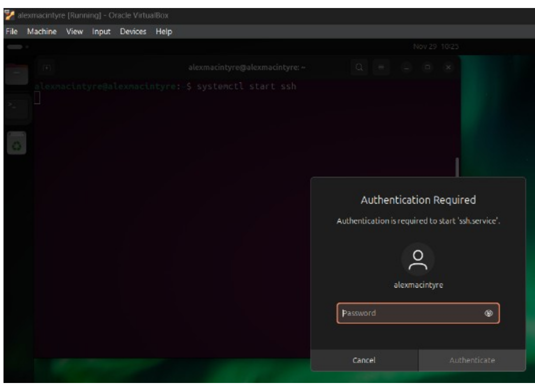




Step Three – Install OpenSSH by typing "**sudo apt install openssh-server**" and pressing **Enter**. There will be a prompt to continue. To continue, type "**Y**" and press **Enter**.

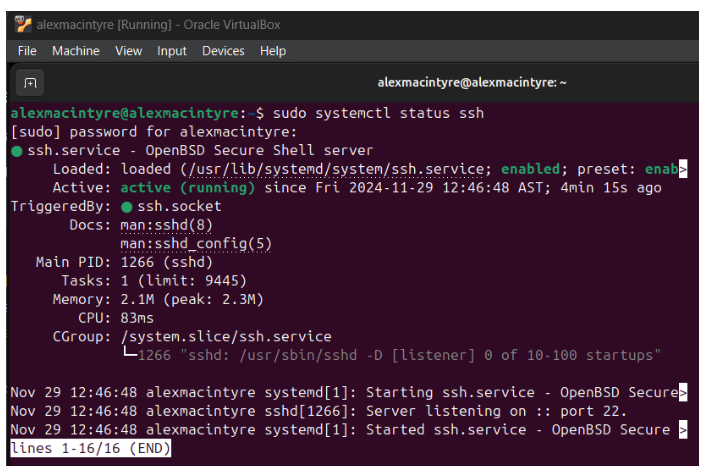
Step Four – Enable SSH by typing "**systemctl enable ssh**" and pressing **Enter**. A dialogue will appear asking for authentication. Enter the sudo password and click "**Authenticate**" repeatedly until the dialogue disappears.





Step Five – Start SSH by typing "**systemctl start ssh**" and press **Enter**. A dialogue will appear asking for authentication. Enter the sudo password and click "**Authenticate.**"

Step Six – To view the status of SSH, type "sudo systemctl status ssh" and press Enter. It should say "active (running)".

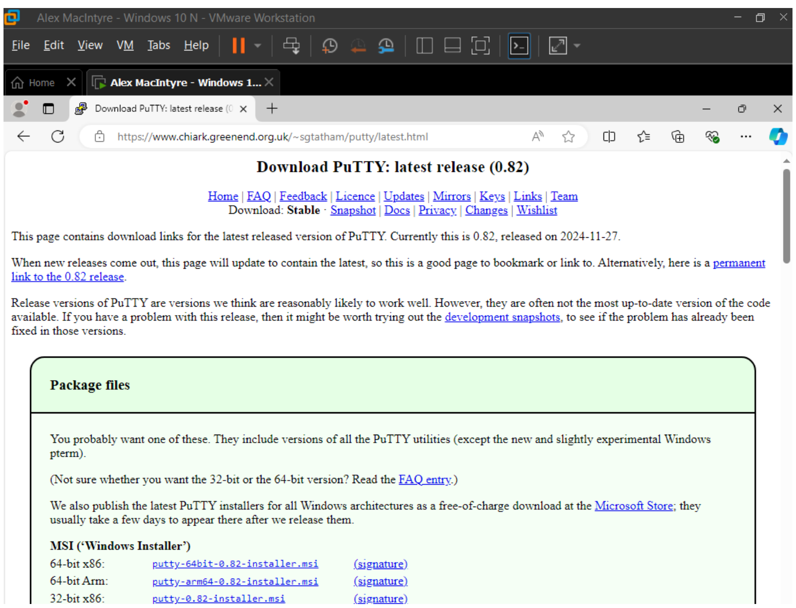


Installing PuTTY on Windows

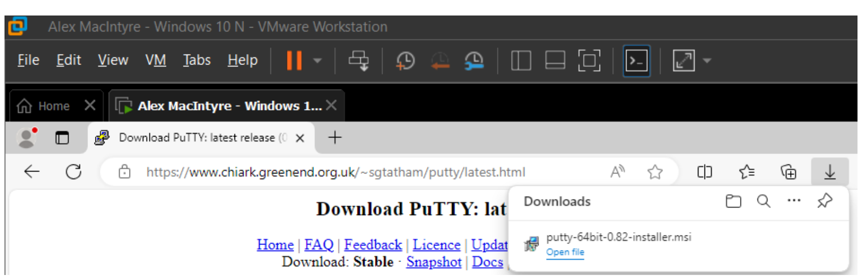
Step One – Open a web browser and enter the following address:

https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html

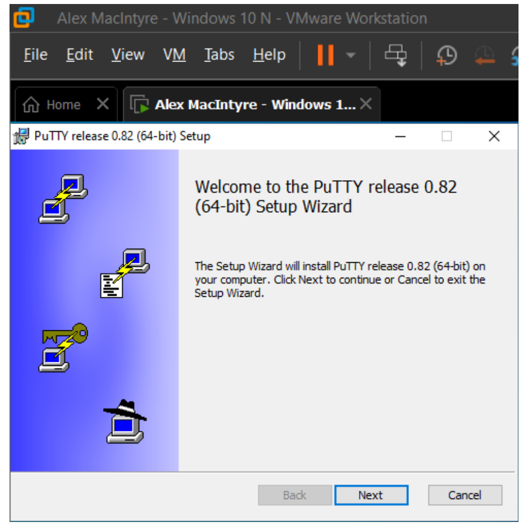
Step Two – Click on the link that matches the system's architecture. In this case, it would be "**64-bit x86**".



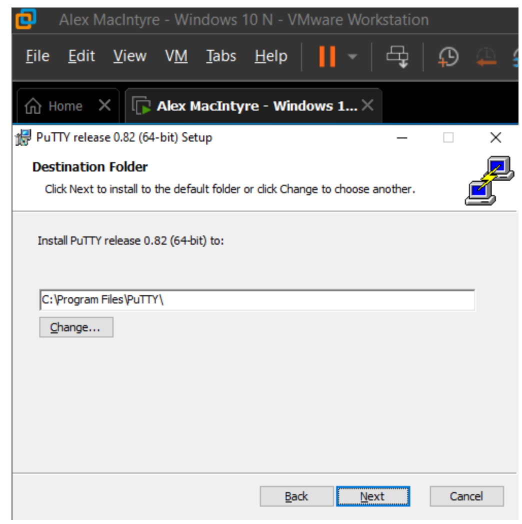
Step Three – Click "**Open File**" once the download has been completed to begin the installation process.



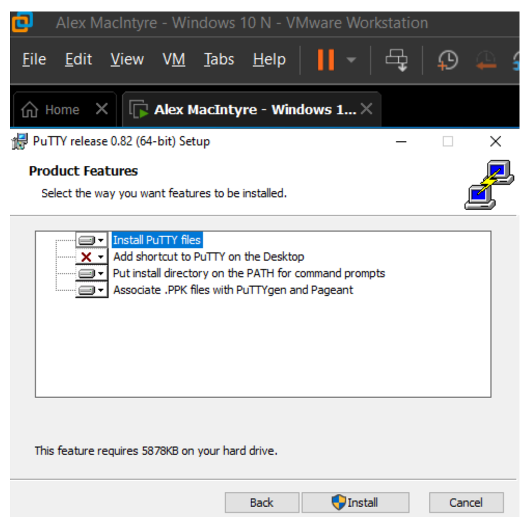
Step Four – Click "**Next**".



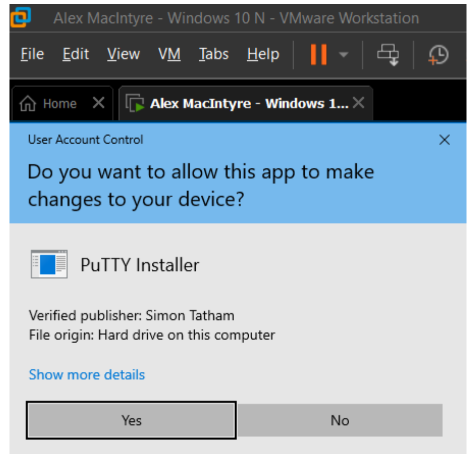
Step Five – Click "**Next**".



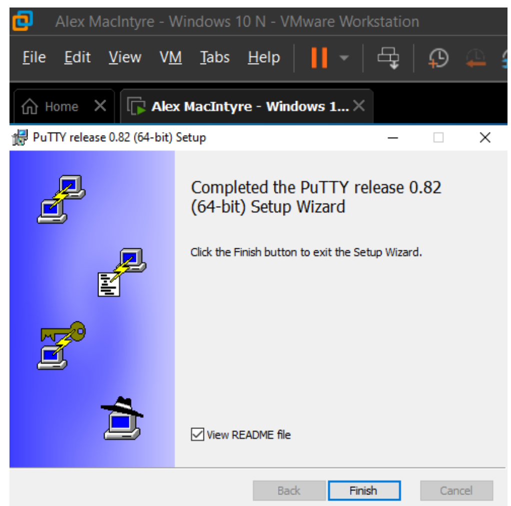
Step Six – Click "**Install**".



Step Seven – Allow the installer to make changes. Click "**Yes**".

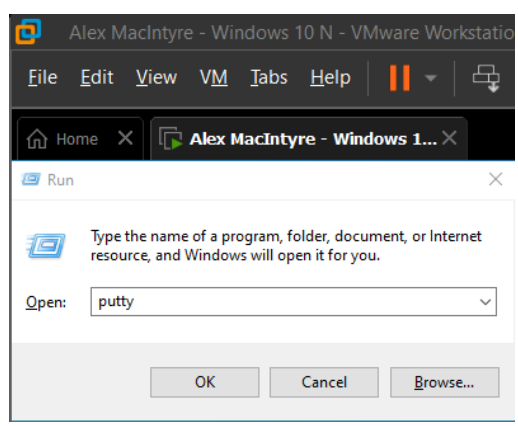


Step Eight – Click "**Finish**".

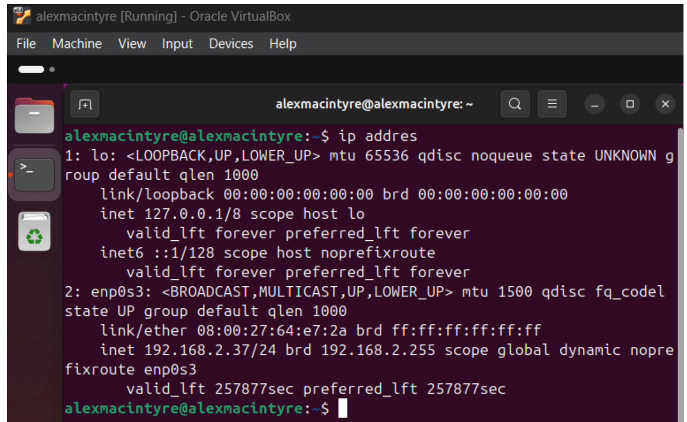


Creating a Connection

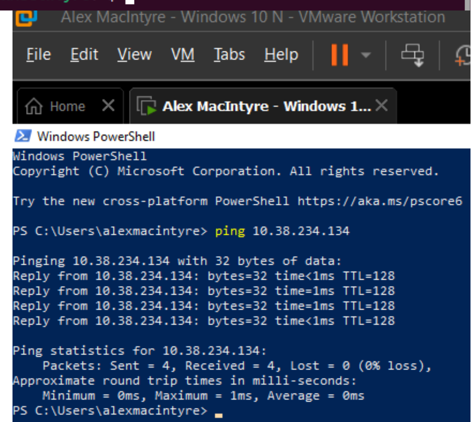
Step One – Open PuTTY on Windows. It can be done by pressing **Windows + R**, typing "**PuTTY**," and pressing "**OK**". This can also be done by pressing the **Windows** key and searching for "**PuTTY**."



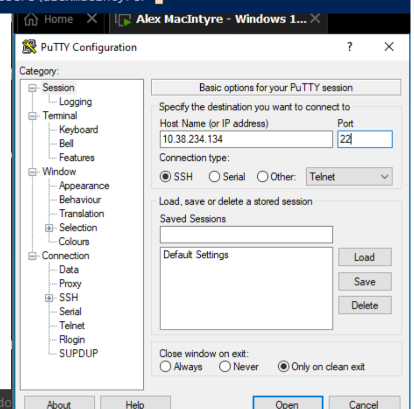
Step Two – In the Ubuntu machine, verify the IP address by typing "**ip address**" into the terminal and pressing Enter. The correct IP address will be under the section beginning with "**2: enp0s3**" and in the "**inet**" line. Here, the IP address is "**192.168.2.37/24**".



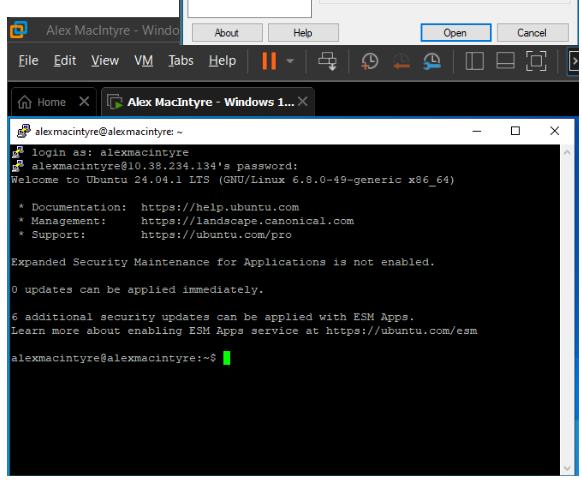
Step Three – On the Windows machine, open PowerShell by pressing "**Windows + X**" and then "**A.**" Ping the Ubuntu machine by typing "**ping**" followed by the IP address assigned to the other ma. Here, the command is "**ping 192.168.2.37**".



Step Four – Now that the Ubuntu IP has been pinged, and it is known here as "**192.168.2.37,**" it can be entered into the "**Host Name (or IP address)**" box in the PuTTY Configuration. Press "**Open**" to begin the connection.

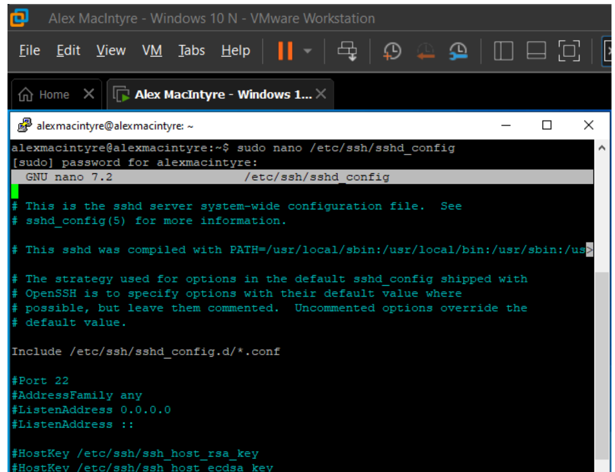


Step Five – Once opened, the terminal will ask for the **sudo username and password**. Input them, and press **Enter** each time. The connection between the two PCs will now be complete.

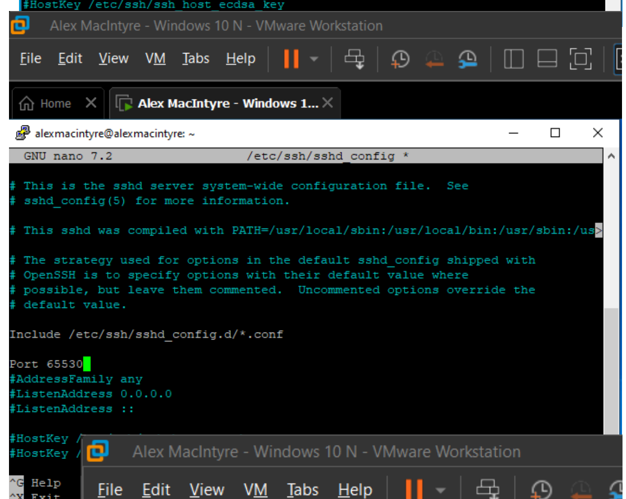


Changing the SSH Port

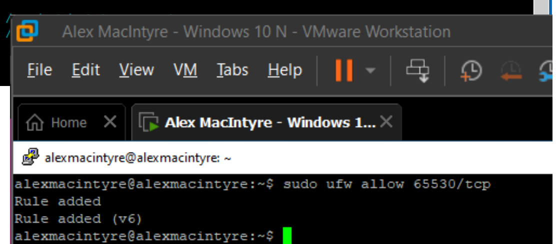
Step One – In the PuTTY terminal, type **"sudo nano /etc/ssh/sshd_config"** and press **Enter.** Input the **sudo password** and press **Enter** again.



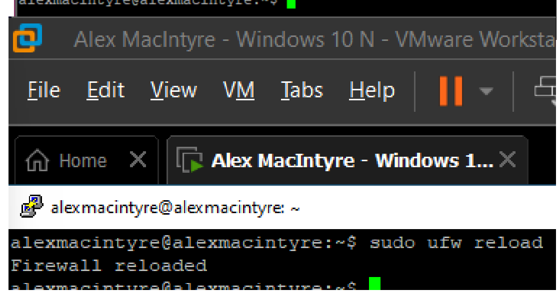
Step Two – To change the port, remove the "**#**" comment indicator from the start of the line that says "**#Port 22**". Change the "**22**" to any port number that isn't identified in a "**ss -ant**" scan, as shown in **Part I – Step Twelve**. Here, it will be "**65530**". Press "**CTRL+X**" to exit the Text Editor and "**Y**" to save the modification.



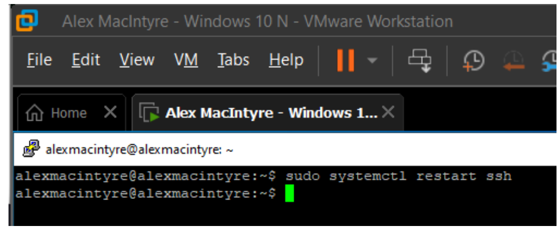
Step Three – Type "**sudo ufw allow 65530/tcp**" to allow the new port through the firewall. Press **Enter**.



Step Four – Type "**sudo ufw reload**" to reload the firewall. Press **Enter**.

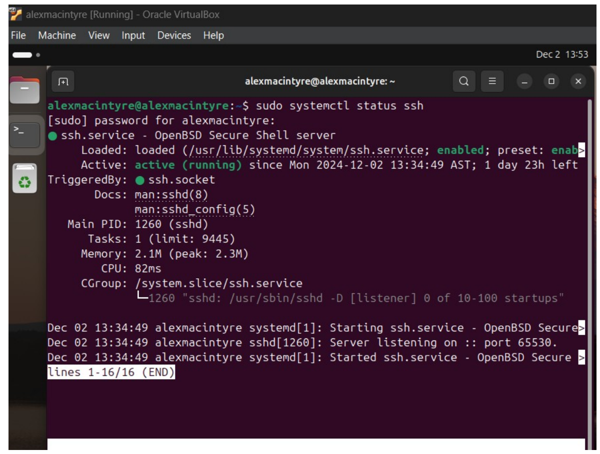


Step Five – For these changes to reflect, type "**sudo systemctl restart ssh**" and press **Enter**.

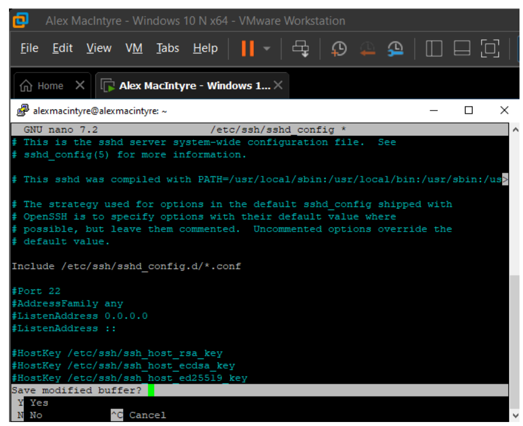


Step Six - After restarting the SSH service, type "**sudo systemctl status ssh**" and press **Enter** to ensure that the listening port is set to the newly established port. It will appear on the second dated line.

"Server listening on :: **port 65530**"

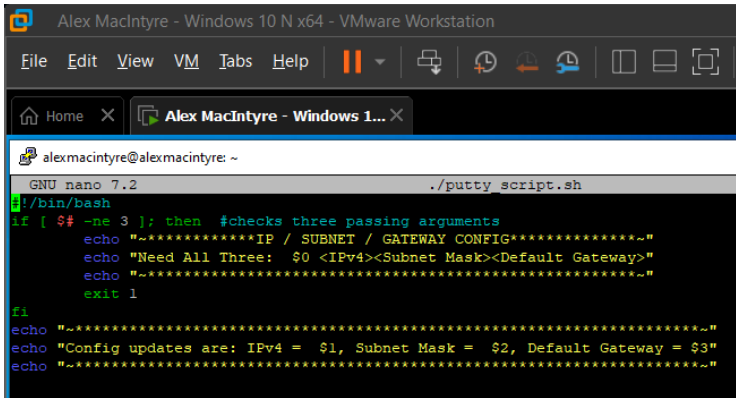


Step Seven – To change the listening port back to Port 22, replace Port 65530 in the text file with Port 22. Press **CTRL+X** and then **"Y"** to save the file. Reload and restart the service with "**sudo ufw reload**" and "**sudo systemctl restart ssh.**"



Scripting with PuTTY

Script Code



Syntax:

if [$# -ne 3]; then #checks three passing arguments

echo “~*************IP / SUBNET / GATEWAY CONFIG***************~”**

echo “Need All Three: $0 <IPv4><Subnet Mask><Default Gateway>”

echo “~*~”**

exit 1

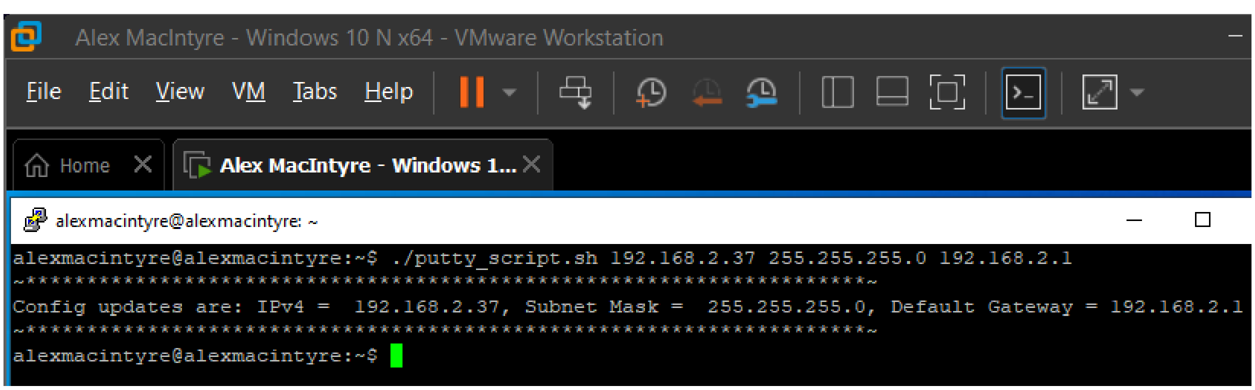
fi

echo “~*~”**

echo “Config updates are: IPv4 = $1, Subnet Mask = $2, Default Gateway = $3”

echo “~*~”**

Working Script



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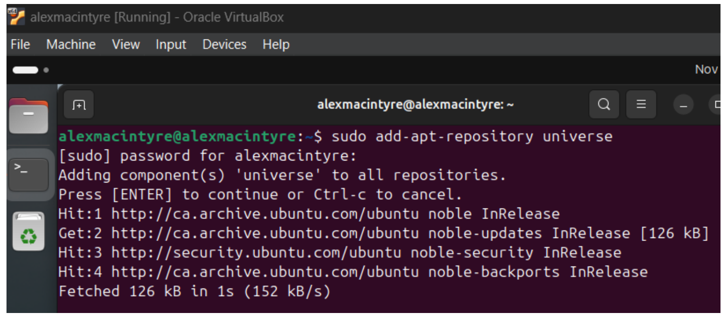
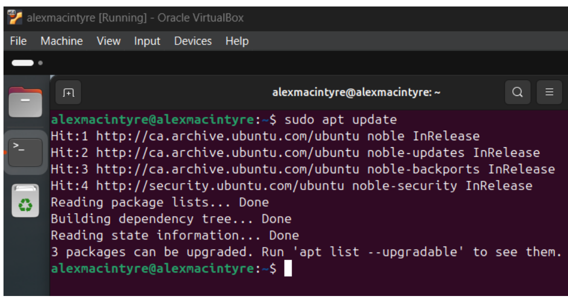
To show that it was completed successfully, here is a PuTTY installation process on Ubuntu:

Installing PuTTY on Linux

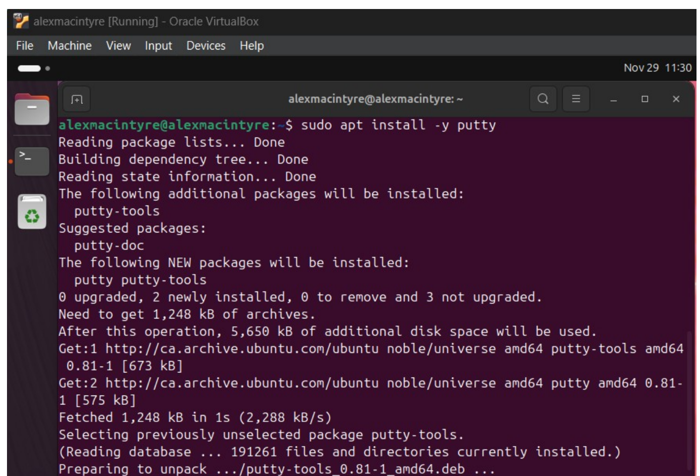
Step One – To install PuTTY, the Universe Repository will need to be added. Type "**sudo add-apt-repository universe**" and press **Enter**. There will be a prompt to continue. To continue, press **Enter**.



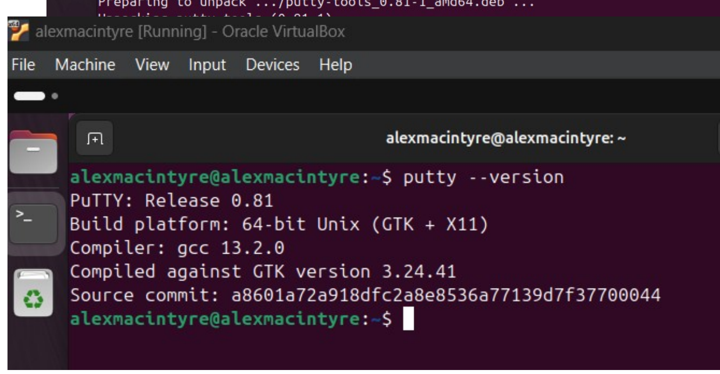


Step Two – Since a new repository was added, there will need to be an update. Type "**sudo apt update**" and press **Enter**.

Step Three – Install PuTTY by typing "**sudo apt install -y putty**" and pressing **Enter**.



Step Four – Verify that PuTTY was successfully installed by typing "**putty –version**" and pressing **Enter.**



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